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① Genetic Optimisation Algorithm

Pseudocode :-

Step1:- Define population for the test

Step2:- Check each individual against a fitness parameter

Step 3:- Loop as long as condition is satisfied

Step 3.1:- Select the best individual as the parent

Step 3.2:- Perform crossover to create new offsprings

Step 3.3:- Perform mutation to introduce small changes

Step 3.4:- Evaluate the fitness of the new offspring

Step 3.5:- Create new population by replacing old population with new offspring

Step 4:- Return best individual

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CPU task Scheduling

*Task which usually need less resources end up waiting till others bring in resources which are not in CPU.

*Here through the soln we try to optimise the problem by scheduling task acc to the resources needed

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CPU task Scheduling

Step 1:- Get all the tasks that need to scheduled

Step 2:- ~~Sort them according to the tasks which need less resource & have the most resource in the register~~

Step 2:- loop until a condition is satisfied

condition :- No more task left to schedule

Step 2.1:- Select the tasks which need the least resource & have the most

Step 2.2:- If multiple required resource in CPU

Step 2.2 :- If multiple tasks fullfill the criteria. Schedule them one after another

Step 2.3:- Remove those tasks which are scheduled from the population

Step 3:- Return the new tasks scheduling scheme

OUTPUT

Task 1 → CPU 2
Task 2 → CPU 3
Task 3 → CPU 1
Task 4 → CPU 2
Task 5 → CPU 1
Task 6 → CPU 2
Task 7 → CPU 1
Task 8 → CPU 3
CPU 1 → 15
CPU 2 → 14
CPU 3 → 18
Minimum Maximum Execution Time : 15